

Low-Poly Vector 3D Modeling Tool

System Architecture & Technical Specification Documentation

1. Vision & Core Philosophy

This project aims to bridge the gap between 2D vector graphic design (specifically Inkscape's node-based workflows) and low-poly 3D modeling[cite: 5].

Core Tenets

- **No Traditional Mesh Editing:** Bypasses traditional vertex-edge-face extrusion and box-modeling paradigms[cite: 5].
 - **Vector-Driven Surface Generation:** Models are defined and modified by drawing, lofting, revolving, and combining 2D/3D Bezier vector curves[cite: 5].
 - **Low-Poly Procedural Native:** Geometry is procedurally generated from parametric profiles and curves, optimized for clean `.obj` exports[cite: 5].
 - **Non-Destructive & Hierarchical:** Shapes remain editable via their underlying vector paths throughout the entire design process[cite: 5].
 - **Accessibility & High Precision:** Focuses on spacious 2D vector precision without cluttering the screen with tiny multi-pane viewports.
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2. Toolset Architecture

2.1 General Tools

- **Selection Tool:** Selects, translates, rotates, and scales whole objects/sub-meshes in 3D space[cite: 5].

2.2 Path & Lofting Tools

- **Edit by Nodes (Vertex & Edge Tool):** Inkscape-style node editing for 2D and 3D paths[cite: 5]. Supports inserting nodes between selected vertices, deleting nodes, adjusting handle tangents, and snapping[cite: 5].
- **3D Bezier Pen Tool:** Draws vector paths in 3D space across orthographic or perspective planes with 3D Bezier control handles[cite: 5].

- **Cross-Section Sweeping ("Stroke-to-Solid"):** Assigns a 2D profile (e.g., triangular tube, square tube, flat ribbon) along a Bezier path with an adjustable **Segment Count** slider (3–8 sides) to generate volumetric stroke geometry dynamically[cite: 5].
- **Revolve (Lathe) Tool:** Generates rotational 3D solids by sweeping a half-profile 2D vector sketch around a designated axis (e.g., Y -axis centerline). Features parametric controls for **Sweep Angle** (0° – 360°), **Radial Segment Steps** (e.g., 3–32 steps for low-poly faceting), and **Axis Offset**. Ideal for symmetrical objects like vases, cups, barrels, domes, and wheels.

2.3 Solid Primitive Tools

Tools that directly generate volumetric geometry:

- **Rectangular Prism:** Defined parametrically by **Width**, **Height**, and **Depth**[cite: 5].
- **Sphere:** Defined parametrically by **Radius** and base tessellation steps[cite: 5].
- **Pyramid:** Defined parametrically by **Base Shape** (*Circle, Rectangle, or Equilateral Triangle*), **Base Area**, and **Height**[cite: 5].
- **Platonic Solids Tool:** Parametrically generates regular convex polyhedra by selecting **Type** (*Tetrahedron, Cube/Hexahedron, Octahedron, Dodecahedron, or Icosahedron*) and **Scale/Radius**.
- **Trapezohedron Tool:** Parametrically generates n -gonal trapezohedra (duals of antiprisms) defined by **Degree** (n -gon base, e.g., trigonal, tetragonal, pentagonal), **Radius**, and **Height/Asymmetry**.

3. Viewport Navigation & Canvas Interface

Designed to maximize screen real estate for precise vector node editing while allowing full 3D camera exploration:

3.1 Hybrid Viewport Modes

- **Single Viewport Default:** The default layout presents a single, spacious workspace maximizing real estate for node editing.
- **Picture-in-Picture (PiP) Overlay:** A small, resizable corner preview window displays the live 3D rendered volume while drawing on the main 2D orthographic canvas. Clicking the PiP window swaps the 2D and 3D views instantly.
- **Optional Viewport Splitting:** Supports side-by-side split screens with adjustable, thick drag-and-drop splitter bars.

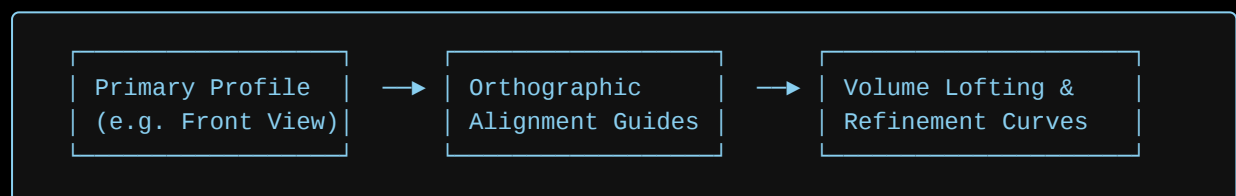
3.2 Camera Control & View Lock System

- **Free 3D Orbit:** Users can freely orbit, tilt, pan, and zoom around the model in 3D space.

- **Orthographic Plane Snap:** Clicking an axis on the Orientation Gizmo snaps the camera flat to cardinal planes (Front, Left, Top).
 - **Universal View Lock Toggle:** A quick UI toggle button (and hotkey) that completely freezes camera rotation at **any arbitrary angle or perspective view** (not restricted to orthographic planes). When locked, navigation inputs on the canvas are restricted to 2D panning and vector node editing, preventing accidental camera rotation during fine drawing operations.
 - **Transition Style Settings:** Offers user-selectable camera transitions between **Smooth Orbit Interpolation** and **Instant Cut Snap**.
 - **Infinite Vector Zoom & Accessibility:** Zooming in close maintains crisp, resolution-independent vector lines and enlarged node control handles for high visual accessibility.
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4. Orthographic Lofting & Mesh Generation Workflow

Generating complex 3D forms from 2D profile paths relies on an orthographic cross-section lofting pipeline[cite: 5]:



4.1 Primary Contour & Depth Profiling

1. **First Profile:** User draws a 2D closed/open profile on a cardinal view (e.g., Front View)[cite: 5].
2. **Depth Profile:** User switches to a perpendicular view (e.g., Side/Left View) and draws depth-defining curves[cite: 5].
3. **Volume Triggering:** Volume can be generated automatically upon detecting a valid curve set, or manually via a keyboard shortcut/UI button[cite: 5].

4.2 Automatic Orthographic Bounding & Alignment Snapping

To prevent topological gaps and non-manifold geometry[cite: 5]:

- **Projection Guides:** Switching views automatically projects thin, ghosted reference lines and bounding extents of the primary profile[cite: 5].
 - **Node Snapping:** Start and end points of depth curves snap directly to projected boundary markers, ensuring exact alignment across axes[cite: 5].
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5. SVG Layer Parser & Automatic Assembly

5.1 Structural Mapping & Layer Extraction

The importer parses SVG document tree structures (`<g>` groups and nested layers) and maps them directly into a 3D Scene Graph parenting hierarchy:

SVG Group Hierarchy	3D Scene Graph
├─ Layer 1: Torso	→ Root Mesh (Torso)
└─ Sub-layer: Head	→ Child Mesh (Head, parented to Torso)
└─ Sub-layer: Ears	→ Grandchild Mesh (Ears, parented to Head)
└─ Sub-layer: Muzzle	→ Grandchild Mesh (Muzzle, parented to Head)
└─ Sub-layer: Near_Leg	→ Child Mesh (Near_Leg, parented to Torso)

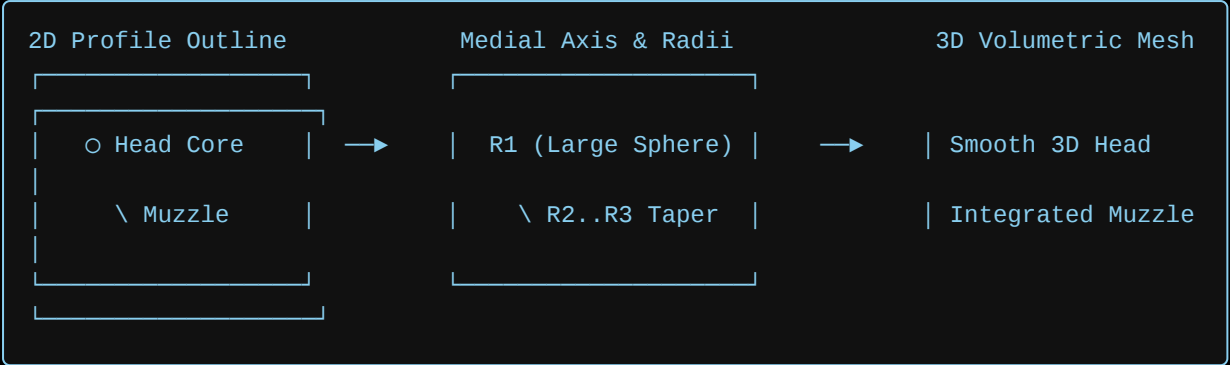
- **Global Coordinate Preservation:** Vector paths retain their shared 2D canvas coordinates (X, Y) across all layers upon extraction, preventing manual 3D re-assembly.
- **Layer Stacking to Z-Depth Mapping:** SVG rendering layer order (top-to-bottom) maps directly to default initial 3D Z-depth offsets ($+Z$ for front layers, $-Z$ for back layers).
- **Surface Socket Auto-Generation:** Overlapping junction points automatically initialize Surface Socket Pivots at parent-child intersection centroids.

6. Volumetric Inflation Algorithms

6.1 Medial Axis Sphere Inflation (Cranium & Muzzle Topology)

For organic geometry with varying regional volume (such as a round head connected to a tapering snout), the engine utilizes **Medial Axis (Maximal Inscribed Circle) Inflation**:

1. **Skeleton Extraction:** Computes the interior Medial Axis Transform (MAT) from the closed 2D boundary path.
2. **Inscribed Radius Sampling:** Calculates the maximum inscribed radius $R(t)$ touching the contour boundary at every point along the axis.
3. **Implicit Surface Sweeping:** Generates a smooth 3D isosurface skin around the sphere field defined by $R(t)$ (large spherical skull core tapering continuously into the muzzle).



6.2 Dual-Axis Sweep (Paws & Tapered Limbs)

- 1. **Centerline Spine Extraction:** Derives a 1D skeleton curve along the longitudinal axis of the 2D limb sketch.
- 2. **Variable Elliptical Cross-Sectioning:** At each point s along the spine, an elliptical slice is evaluated:
$$\text{Radius}_Z(s) = \text{Distance from spine to Side Profile boundary at } s$$
$$\text{Radius}_X(s) = \begin{cases} \text{Radius}_Z(s) \times k & \text{if Mode = Derived} \\ \text{Distance to Front Profile at } s & \text{if Mode = Custom Sketch} \end{cases}$$
- 3. **Skinning:** Sweeps continuous quad/triangle topology across all elliptical cross-sections to generate clean limb and paw geometry.

7. Profile Binding Channels & Multi-Axis Geometry

7.1 Non-Destructive Profile Binding Architecture

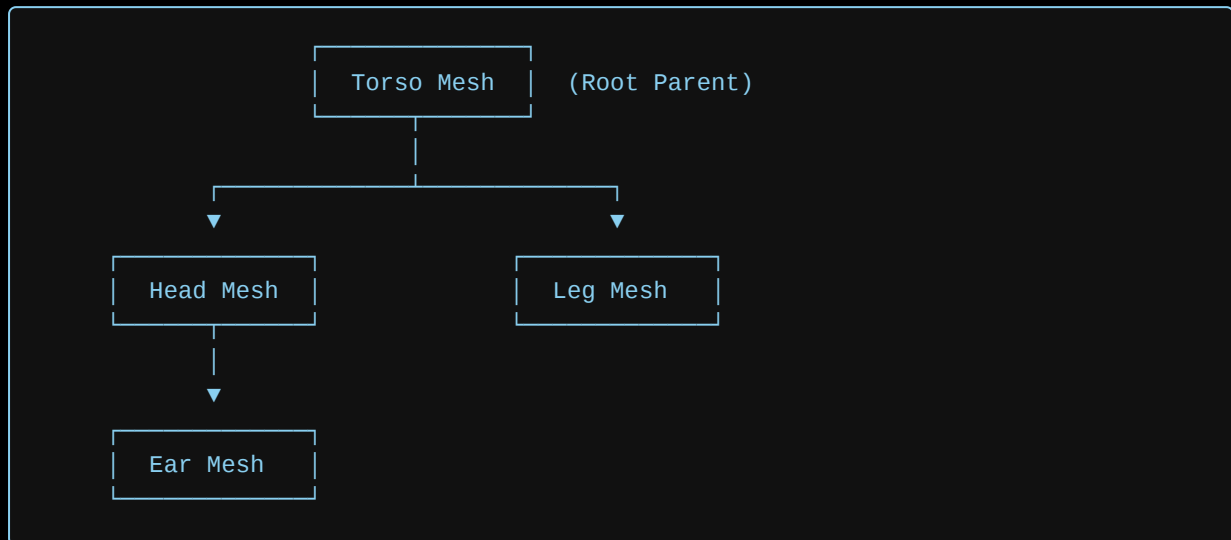
To support 1-click conversion while enabling multi-view geometric control, each sub-component exposes three independent **Profile Binding Channels**: X -Width, Y -Height, and Z -Depth.

7.2 Binding Modes per Channel

- 1. **Follow Sketch [Path_ID] (Primary Default):** Directly bound to the active 2D vector path drawn or imported in that view projection.
- 2. **Derived (Zero-Error Fallback):** Automatically computes profile thickness along the orthogonal axis based on the primary sketch dimensions ($\text{Radius}_X = \text{Radius}_Z \times \text{Width Multiplier}$). Guarantees immediate 3D volume without producing flat or zero-thickness errors.
- 3. **Custom Sketch (Manual Override):** Replaces the **Derived** fallback with explicit 2D vector data drawn on a perpendicular reference plane (e.g., Front View).

8. Multi-Shape Hierarchy & Assembly

Complex subjects (e.g., character models) are built by assembling simpler component sub-meshes rather than stitching a single complex outline[cite: 5].



8.1 Component-Based Grouping

- Shapes are broken down into self-contained sub-objects (e.g., *Head, Torso, Limbs*)[cite: 5].
- Sub-shapes overlap naturally in 3D space without requiring continuous skin-stitching algorithms[cite: 5].

8.2 Surface-Anchored Joint Sockets

- **Root Profile Tapering:** Child sub-shapes automatically taper slightly at their connection rim, forming a clean joint transition instead of harsh geometric clipping[cite: 5].
- **Z-Fighting Prevention:** A minute scale/depth offset (1–2%) is applied at connection planes to eliminate flickering textures[cite: 5].

9. Parenting, Surface Locking & Collision Rules

To maintain logical placement during manipulation and prevent clipping errors[cite: 5]:

9.1 Parent-Child Linking & Sockets

- Users select elements and assign relationships (e.g., *Ear → Head → Torso*)[cite: 5].
- **Automatic Socket Pivots:** Linking automatically places the child's rotation pivot at the intersection point with the parent's surface, aligned to the surface normal[cite: 5].

9.2 Hierarchy-Aware Raycasting

- **Direct Parent Magnetism:** Raycasting for surface attachment only registers the designated direct parent mesh, ignoring unrelated sub-shapes[cite: 5].
- **Collision Filters:** Paws, tails, and opposing limbs do not stick to or sink into each other[cite: 5].
- **Bounding Volume Bumping:** Unrelated sub-objects utilize simplified bounding spheres to prevent physical clipping through adjacent limbs[cite: 5].

9.3 Surface Lock & Parametric Adaptability (UV Surface-Binding)

- **Relative Coordinate Binding:** Child locations are stored as normalized surface coordinates (U, V) relative to the parent's contour paths[cite: 5].
- **Dynamic Morph Adaptability:** Modifying the parent's base vector nodes automatically shifts attached children along the updated surface, preserving alignment without floating or clipping[cite: 5].

10. Mesh Pipeline & Export Specifications

10.1 Dynamic Mesh State

- **Active Curve Stack:** Refinement curves are managed non-destructively in an ordered stack (Base Profile → Curve A → Curve B)[cite: 5]. Curves can be toggled, reordered, or deleted[cite: 5].
- **Live Parametric Mesh:** Mesh geometry updates procedurally in real-time as underlying vector nodes are edited[cite: 5].

10.2 Export Paradigms (.OBJ Format)

Export Option	Mechanics	Primary Use Case
Modular Sub-Meshes (Default)	Keeps sub-shapes as separate distinct geometries (`o Head`, `o Leg`) inside a single file[cite: 5].	Rigging, animation, game engine imports, easy re-editing[cite: 5].
Fused Boolean Union	Merges overlapping volumes into a single continuous manifold skin, stripping internal faces[cite: 5].	3D printing (watertight requirement), physics colliders, sculpting bases[cite: 5].